

Warm-up

- (1) Use a Taylor series to compute $\lim_{x \rightarrow 0} \frac{e^{x^2} - 1 - x^2}{x^4}$.
- (2) Use a Taylor series to compute $f^{(10)}(0)$ for $f(x) = xe^x$.

Medium

- (1) Use a degree 5 Taylor polynomial to approximate $\arctan(1/2)$.
- (2) Using series we already know, write down a formula for the (infinite) Taylor series for $(1 + 3x)^{2/3}$, and then write down the degree-three polynomial explicitly.
- (3) Use a degree 5 Taylor polynomial to estimate $\sin(.3)$. Give an upper bound for the error in that estimate.
- (4) Use a degree 3 Taylor polynomial to estimate $\sqrt{e}$.
- (5) Use a degree 3 Taylor polynomial to estimate $\sqrt{1.2}$.

Stretch

- (1) Give an upper bound for the error in the estimate of (4).
- (2) The normal distribution (bell curve) with mean 0 and standard deviation 1 is

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

If a process is modeled by a normal distribution, the probability that the outcome falls within a certain range $[a, b]$ is found by integrating the distribution from a to b . Estimate the probability that the outcome falls within one standard deviation of the mean using a degree 5 Taylor polynomial. That is, estimate

$$\int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 2 \int_0^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$